THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The radii of two spheres are in the ratio $1 : 5$. The ratio of their volumes is: 1

(a) $1 : 5$ **(b)** $1 : 125$ **(c)** $1 : 25$ **(d)** $1 : 120$

2. Take $\pi = \frac{22}{7}$. The total surface area of a cone of radius 7 cm and slant height 25 cm is: 1

(a) 704 cm^2 **(b)** 633 cm^2 **(c)** 775 cm^2 **(d)** 846 cm^2

3. A cone has radius 20 cm and height 21 cm. Its slant height is: 1

(a) 28 cm **(b)** 29 cm **(c)** 30 cm **(d)** 31 cm

4. A tank measures 5 m $\times$ 10 m $\times$ 12 m. Its capacity in litres (1 m$^3 = 1000$ L) is: 1

(a) 600 **(b)** 600000 **(c)** 6000000 **(d)** 60000

5. The number of cubes of edge 4 cm that can be cut from a cuboid 16 cm $\times$ 20 cm $\times$ 12 cm is: 1

(a) 59 **(b)** 64 **(c)** 60 **(d)** 61

6. A pyramid has a square base of side 8 cm and height 36 cm. Its volume is: 1

(a) 923 cm^3 **(b)** 691 cm^3 **(c)** 768 cm^3 **(d)** 845 cm^3

7. The radius of a sphere is multiplied by 2. Its surface area becomes ___ times: 1

(a) 8 **(b)** 4 **(c)** 5 **(d)** 2

8. The edge of a cube is multiplied by 3. Its volume becomes ___ times: 1

(a) 3 **(b)** 9 **(c)** 27 **(d)** 28

9. The radii of two spheres are in the ratio $1 : 3$. The ratio of their volumes is: 1

(a) $1 : 3$ **(b)** $1 : 27$ **(c)** $1 : 9$ **(d)** $1 : 24$

10. Take $\pi = \frac{22}{7}$. The total surface area of a cone of radius 31.5 cm and slant height 52.5 cm is: 1

(a) 9148 cm^2 **(b)** 8316 cm^2 **(c)** 7484 cm^2 **(d)** 9981 cm^2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The total surface area of a cube is 1176 cm^2. Find its edge, its volume and the length of its face diagonal in terms of $\sqrt{2}$. 2

12. How many cubes of edge 5 cm can be cut from a cuboid measuring 20 cm × 15 cm × 10 cm? Find the leftover volume if each cube is cut and the cuboid is fully used. 2

13. Take $\pi = \frac{22}{7}$. A cylinder of height 8 cm has volume 11088 cm^3. Find its radius and its curved surface area. 2

14. The total surface area of a cube is 96 cm^2. Find its edge, its volume and the length of its face diagonal in terms of $\sqrt{2}$. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Take $\pi = \frac{22}{7}$. A cone has radius 42 cm and height 56 cm. Find its slant height, curved surface area and volume. 3

16. The edge of a cube is 6 cm. If every edge is multiplied by 2, find the new surface area and volume and the factors by which they change. 3

17. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area of a solid hemisphere of radius 14 cm. 3

OR

Take $\pi = \frac{22}{7}$. A cone has radius 7 cm and height 24 cm. Find its slant height, curved surface area and volume. 3

18. The edge of a cube is 8 cm. If every edge is multiplied by 3, find the new surface area and volume and the factors by which they change. 3

OR

Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area of a solid hemisphere of radius 3.5 cm. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A cylindrical metal block of radius 7 cm and height 15 cm is melted and recast into cones of the same radius and height 9 cm. How many cones are made? 5

OR

A solid cuboid 175 cm × 25 cm × 25 cm is melted and recast into cubes of edge 5 cm. Find the number of cubes and the total surface area of all the cubes. 5

20. A cylindrical metal block of radius 7 cm and height 24 cm is melted and recast into cones of the same radius and height 2 cm. How many cones are made? 5

OR

A solid cuboid 64 cm × 16 cm × 16 cm is melted and recast into cubes of edge 4 cm. Find the number of cubes and the total surface area of all the cubes. 5

END OF QUESTION PAPER Total: 40 Marks

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